

ABSTRACT

How do you image the large and the small at the same time? We developed new technologies for fast nanoscale imaging of centimeter-scale specimens - including whole mouse brains - with minimal tiling and no physical sectioning.

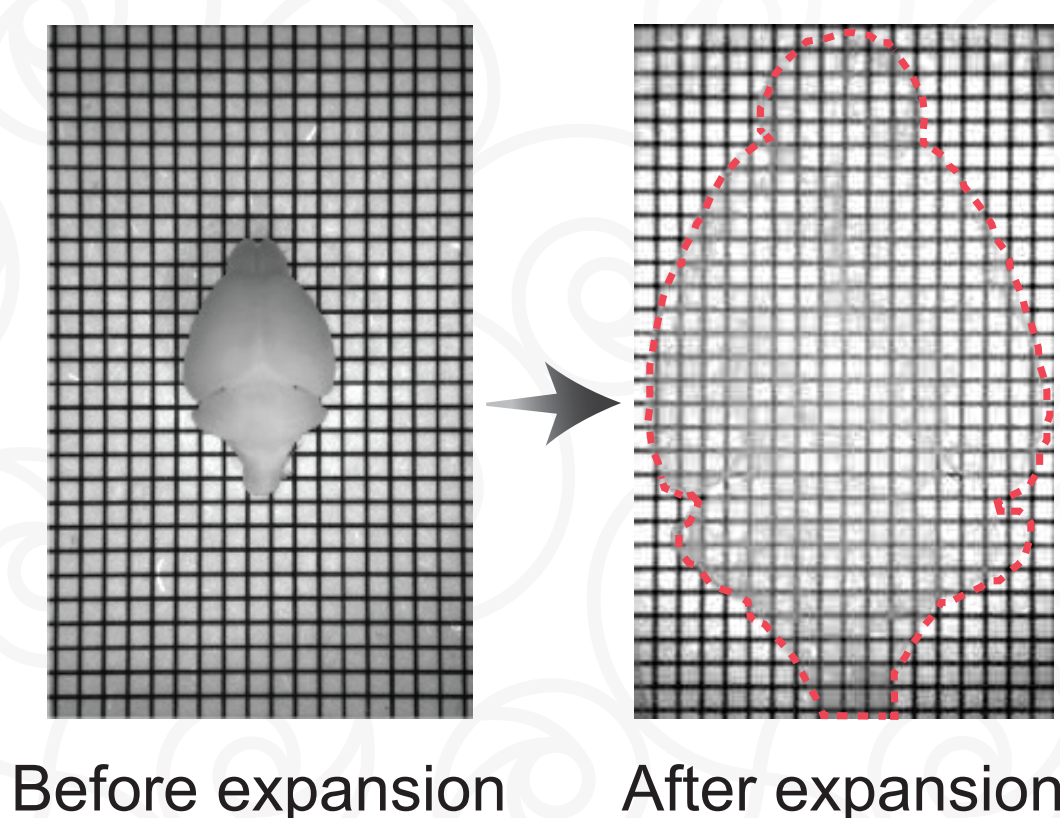
BACKGROUND AND SCOPE

Neurons in the brain have axonal arbors that can be centimeters long but nanometers thin.

Brain mapping and discovering neuron type requires tracing these neurons, but this is difficult with current imaging methods.

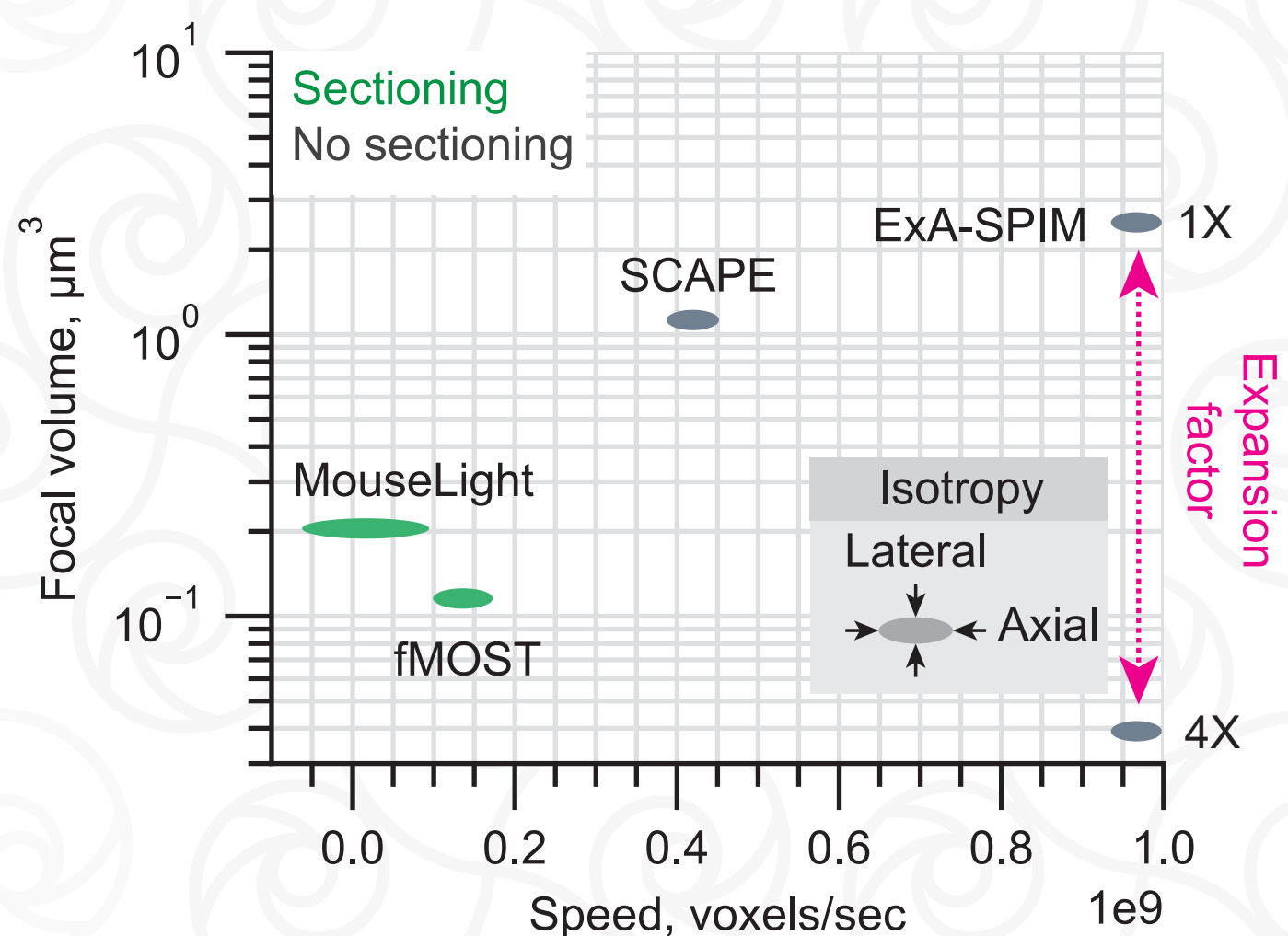
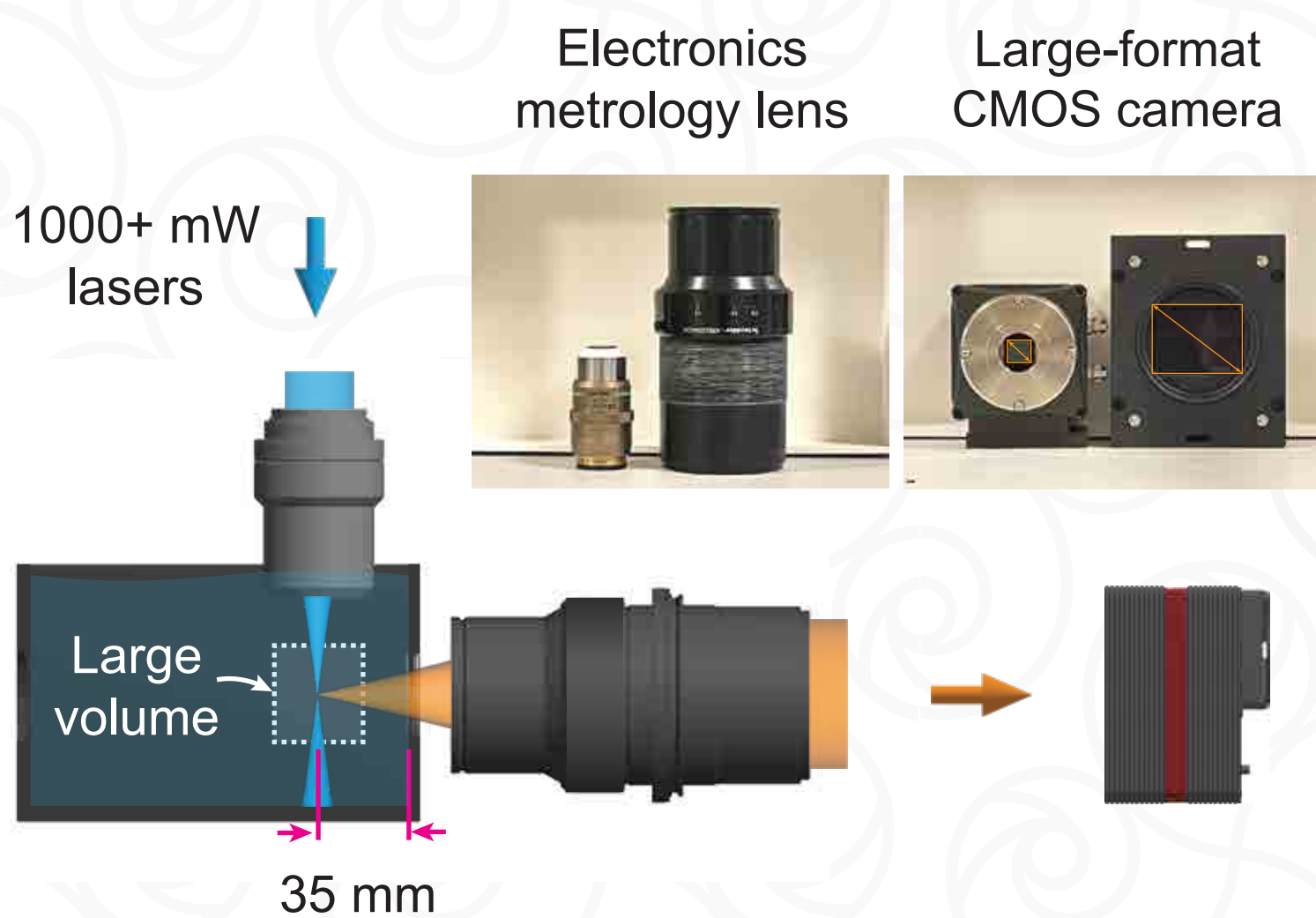
We developed a novel expansion and clearing method for imaging centimeter-scale tissues at nanoscale resolution.

To adapt to imaging these enlarged tissues, we developed the expansion-assisted selective plane illumination microscope (ExA-SPIM)



ExA-SPIM has a field of view ~100× larger (13.3 mm diameter) and a working distance ~10× larger (35 mm) compared to typical biology objectives, while retaining a relatively high numerical aperture (NA) = 0.305. When combined with 3× expansion, the system achieves high isotropy with an effective optical resolution of ~300 nm laterally, and ~800 nm axially. It additionally uses minimal image tiling, no physical sectioning, and imaging speeds of up to 1 gigavoxels/sec.

The first prototype system was effective, but there were several optical limitations and aspects to be reworked. A new version has been designed and is being built and characterized that addresses many of the problems with the prototype. The process of building and preparing the microscope will be thoroughly documented for broad technology dissemination.



MICROSCOPE COMPONENTS

We added a well-corrected custom excitation objective
The sheet-forming cylinder lens was replaced with a custom Powell lens for uniform intensity

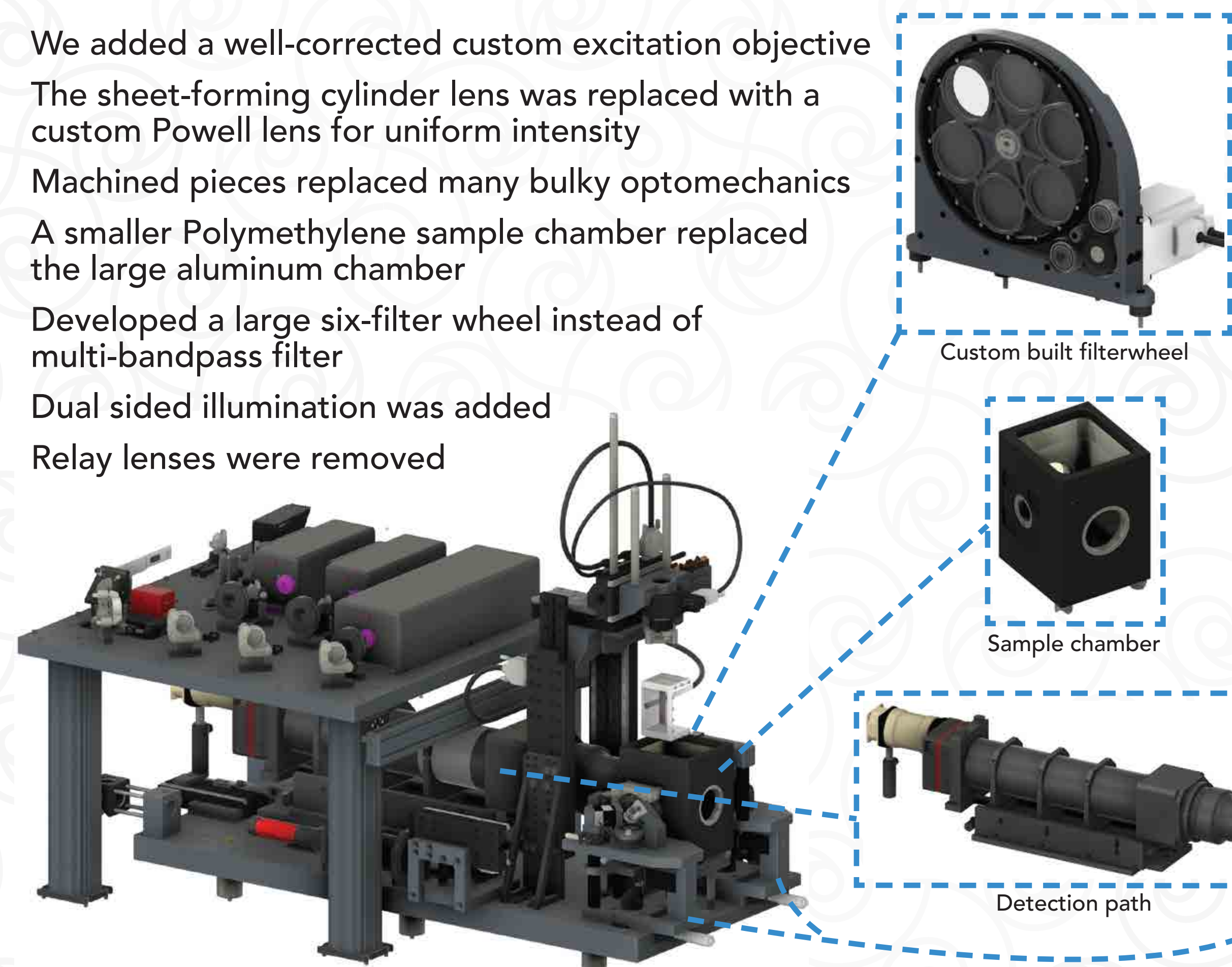
Machined pieces replaced many bulky optomechanics

A smaller Polymethylene sample chamber replaced the large aluminum chamber

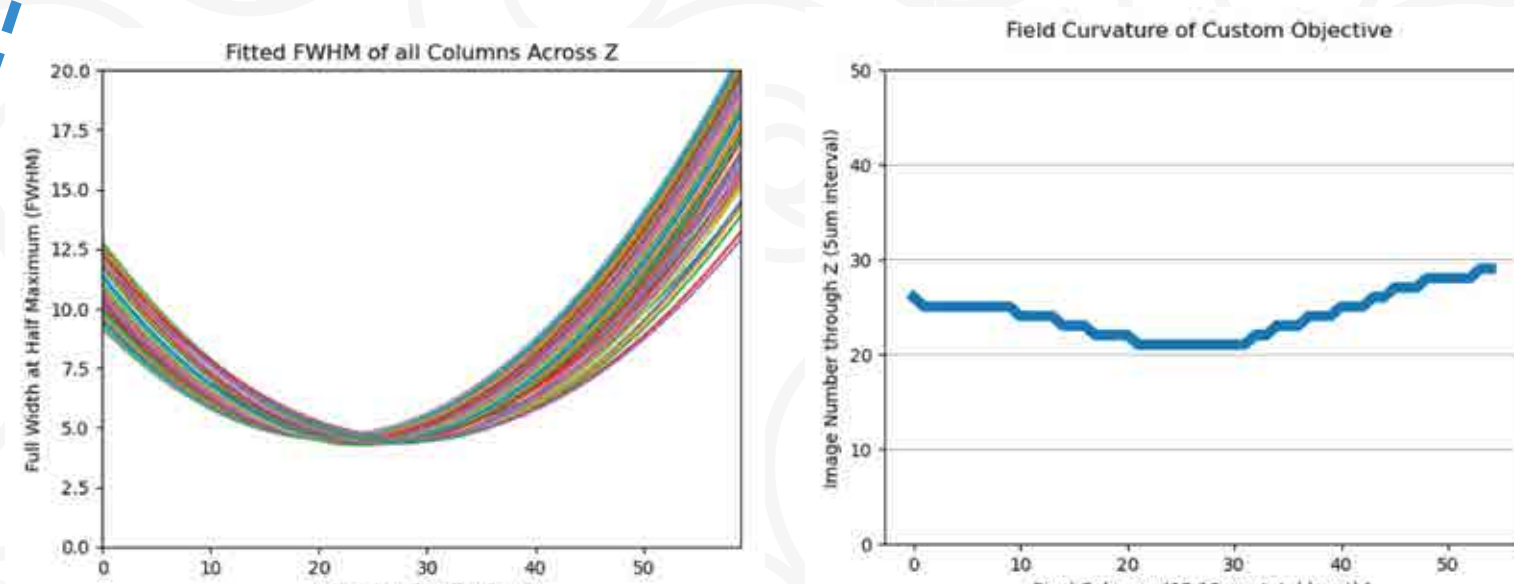
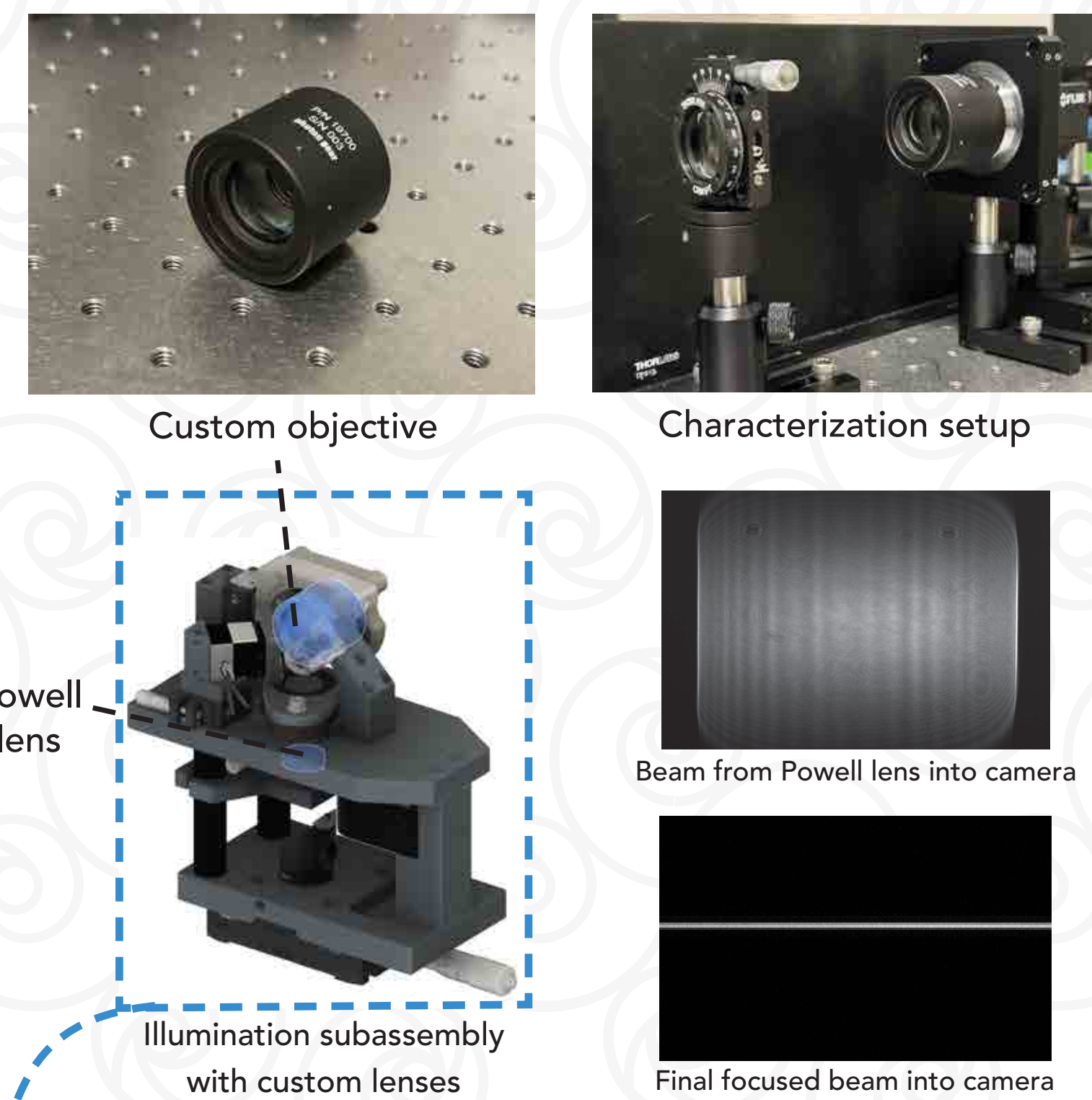
Developed a large six-filter wheel instead of multi-bandpass filter

Dual sided illumination was added

Relay lenses were removed



LENS CHARACTERIZATION



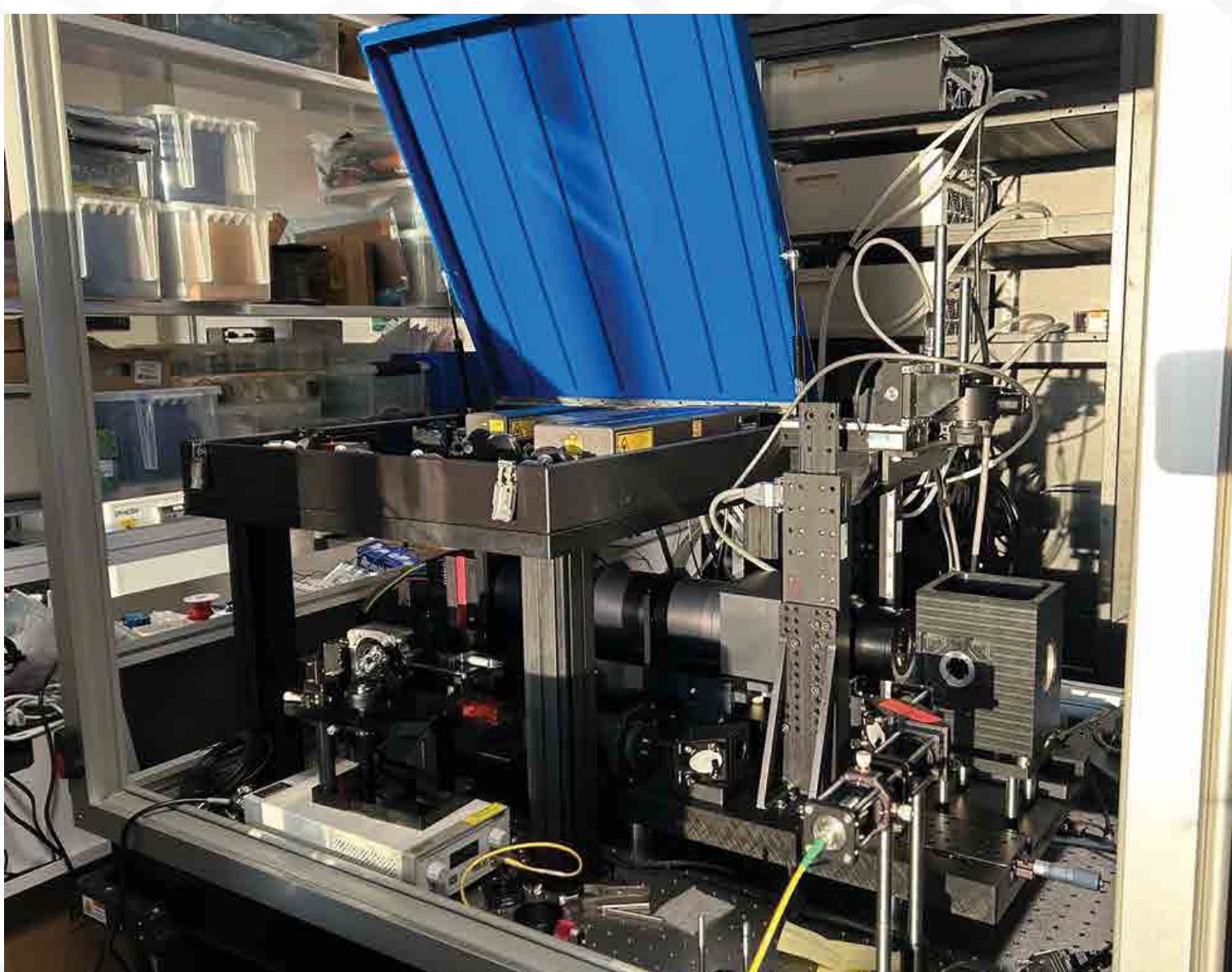
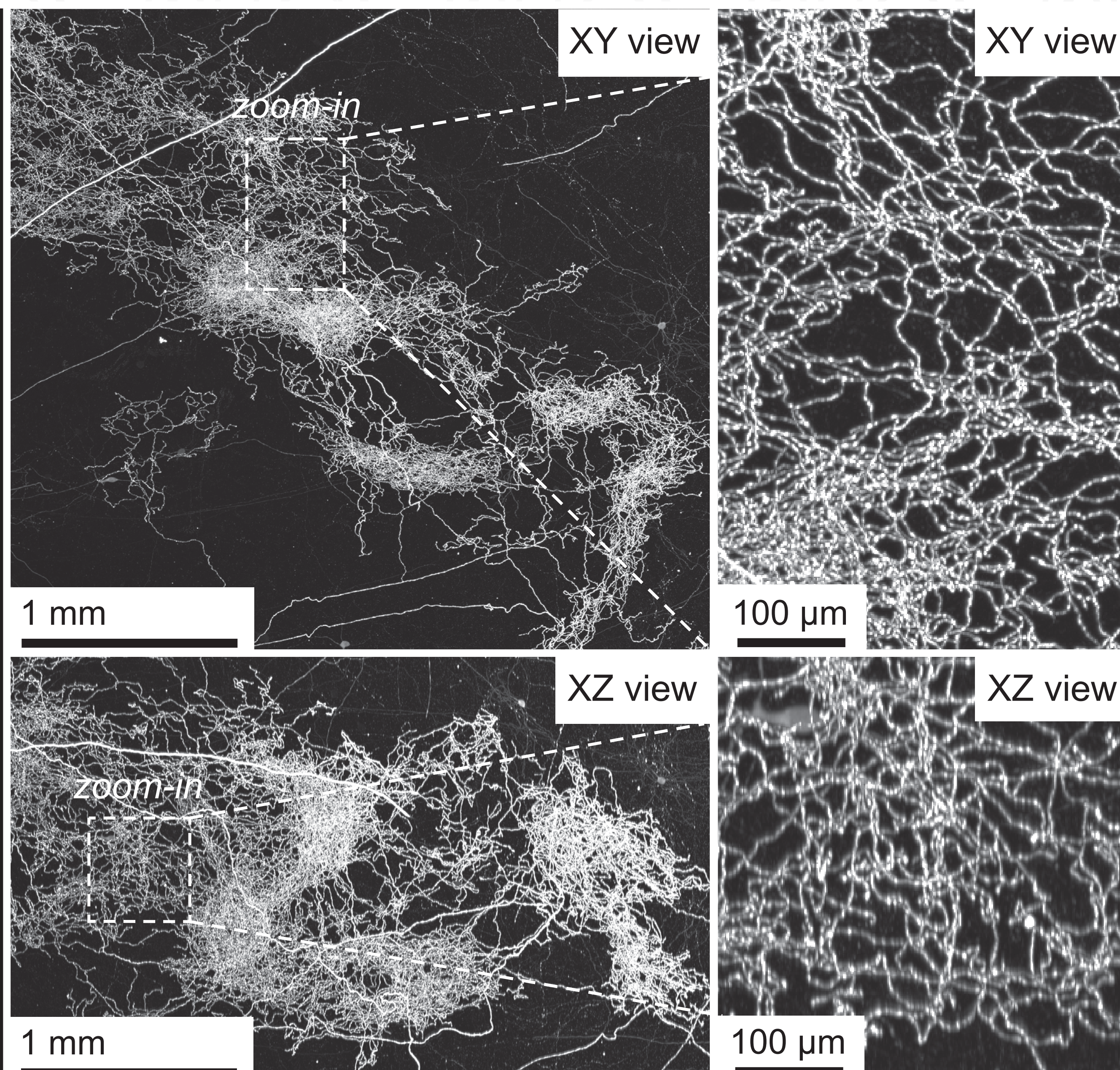
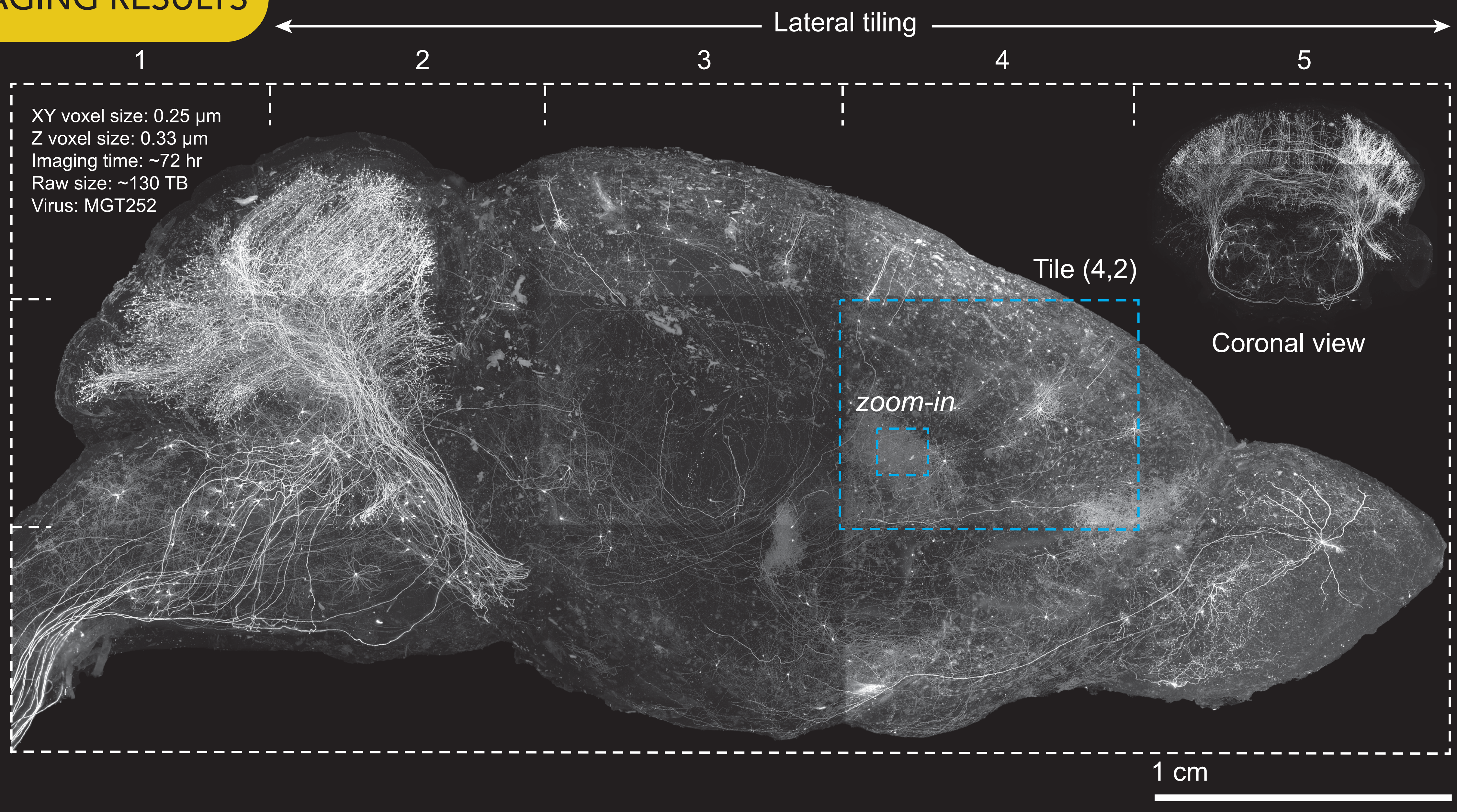
The custom objective forms a focused light-sheet in the imaging plane. Graph shows the width of each part of the beam along Z

The custom objective has low field curvature (40 μm). Graph shows the distance along Z that each part of the beam reaches minimum width (focus)

Graph shows the focused beam intensity profile after Powell lens and objective. Some non-uniformity can be seen, but the profile is more flat than cylinder lenses

The Powell lens fans out columnated light into a more uniform intensity. Graph shows the tested beam intensity profile directly after Powell lens.

IMAGING RESULTS



Read our manuscript for more info

